

AI-Driven Zero-Touch Containerized Network Functions Optimization for 6G Networks Using Network Data Analytics Components

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Abstract—Emerging technologies and use cases, such as telepresence, cobots or massive twinning, bring a new level of demand on performance to future 6G networks. To meet the increasingly strict network requirements of the previously mentioned use cases, networks must go beyond traditional 5G capabilities, enhancing flexibility and resource optimization to support these forthcoming applications, while also reducing the carbon footprint of the virtualized network appliances.

In this paper we investigate the integration of AI-driven Zero-Touch technologies in containerized Virtual Network Functions (VNFs) for future 6G networks. For achieving the presented results we leverage on a Kubernetes-based emulated cluster, a Machine Learning (ML)-enabled Network Data Analytics Function (NWDAF), and both technologies combined to enhance performance and energy efficiency in a Cloud-native and open-source 5G network. With this approach we aim at optimizing network load management in real-time and predicting network behavior to automate resource allocation effectively, thereby reducing human intervention and ensuring robust, scalable network architectures.

Index Terms—6G networks, zero-touch control, NWDAF, AI/ML, cloud native

INTRODUCTION

The advent of 6G networks promises important advancements in wireless communication technologies, emphasizing increased automation, higher data rates, and reduced latency, which are pivotal for supporting the demand for connected devices and IoT applications [1]. As the complexity of network infrastructure increases, traditional network management paradigms become inefficient due to their need for human control, necessitating more dynamic and automated solutions. Virtual Network Functions (VNFs), deployed in containerized environments, offer flexibility and scalability crucial for the modern network demands [2]. These VNFs, orchestrated via Kubernetes, a leading open-source system, enable efficient resource management and scaling in cloud-native environments, which is critical for handling network functions dynamically [3].

In this context, AI-driven Zero-Touch automation technologies, powered by Machine Learning (ML) [4] [5] [6] [7] and integrated with Kubernetes orchestration could offer significant opportunities in enhancing operational efficiencies and reducing manual intervention, pivotal in managing future 6G networks [8]. Kubernetes facilitates automated deployment, scaling, and management of containerized applications, which forms the backbone of modular network functions. On another front, Network Data Analytics Function (NWDAF) under the 3GPP standards for 5G and beyond is an ML-powered component designed to provide network insights that enable predictive analytics and proactive management, enhancing both the service quality and network reliability [9].

In this paper we explore the convergence of Kubernetes orchestration and NWDAF in optimizing containerized VNFs, aiming at a Zero-Touch automation approach suited for the upcoming 6G networks. The synergy between Kubernetes and NWDAF could potentially redefine network management paradigms, paving the way for automated, scalable, and efficient network operations. By harnessing AI to drive automation in VNF management, our proposed Deep Learning (DL) model aims to significantly reduce latency, increase throughput, and enhance the overall network agility, which are essential benchmarks for 6G networks [10]. The integration of these technologies in a unified framework not only streamlines network operations but also ensures robustness against dynamic network changes, embodying the future of telecommunications.

BACKGROUND

In this section we review existing literature on orchestration of containerized VNFs using Kubernetes, the application of ML within NWDAF, and the integration of these technologies in managing and optimizing network functions. As 6G networks emerge, understanding the historical and technological contexts

in which these solutions operate is essential to appreciate their potential impacts and limitations.

Kubernetes in Network Management

Kubernetes has become a cornerstone in the deployment and management of scalable applications in cloud environments. Its role extends into the telecommunications industry, where it supports the orchestration of VNFs—a critical component of modern network infrastructures. Several studies [11] [12] have shown that Kubernetes not only enhances the flexibility of network services but also supports dynamic scaling and recovery, which are crucial for maintaining robust network services. Literature discusses various Kubernetes controllers, such as the Horizontal Pod Autoscaler (HPA) and the Vertical Pod Autoscaler (VPA), that automate the scaling of applications based on observed CPU usage and other metrics [12]. In the telco cloud domain the HPA and VPA methods play an important role in ensuring efficient resource utilization and maintaining service reliability in dynamic network environments. Together, these autoscaling paradigms, which we investigate in the following section, could help telecom operators handle fluctuating workloads, optimize infrastructure costs, and maintain low-latency, high-availability services—key requirements for B5G CNFs.

NWDAF in Anomaly Detection and Resource Management

The implementation of NWDAF in B5G networks represents a significant step towards intelligent network management based on the analysis of compute and cellular performance KPIs. By leveraging on network slicing-focused ML algorithms [13] [14] [15], NWDAF could provide insights that should help in predicting network loads, detecting anomalies, and optimizing resources allocation dynamically [9]. The literature on NWDAF highlights its potential to enhance the Quality of Service (QoS) by adapting network behavior in real-time to changes in traffic patterns and network states [16]. Studies have further elaborated on specific ML models used within NWDAF, such as supervised, unsupervised, and semi-supervised learning algorithms, which facilitate efficient and proactive network management [17].

Synergy Between Kubernetes and NWDAF

Integrating Kubernetes and NWDAF presents a promising area of study, particularly in the orchestration and optimization of network functions. Our research has begun to explore how Kubernetes-based telco-cloud infrastructures scaling can be managed using NWDAF-deployed ML models focused on predicting load anomalies for 5G NFs and slice instances. This synergy aims to reduce manual configurations, enhance network responsiveness, and improve the efficiency of resource utilization in B5G environments.

Our literature review thus contextualizes the importance of Kubernetes and NWDAF in the network sector, particularly as networks evolve towards more automated and intelligent systems. In the following chapters we will build on this foundation, analyzing primary data and experimental results to

evaluate the effectiveness of these technologies in an emulated 5G network environment.

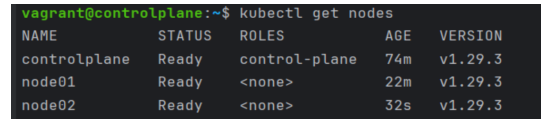
ANALYSIS OF KUBERNETES ORCHESTRATION RESULTS

In this section we look into the results of using Kubernetes orchestration for managing CNFs within our Future Networks 5G SA release 16-compliant pre-commercial testbed featuring a substantial Edge Computing deployment [18]. The emphasis is on evaluating the effectiveness of Kubernetes scaling mechanisms, namely, Horizontal Pod Autoscaler (HPA) and Vertical Pod Autoscaler (VPA), in dynamic network environments.

Horizontal and Vertical Pod Autoscaling

The HPA and VPA are Kubernetes components designed to optimize the allocation of resources in cloud-native environments. The HPA adjusts the number of pod replicas in a deployment based on observed CPU utilization or other select metrics provided by the user. In contrast, the VPA adjusts the CPU and memory limits of pods in a deployment, based on the requirements of the workload.

For our testing activities an emulated Kubernetes cluster was created on top of type-1 hypervisor hosted on our Edge Computing infrastructure, VMware vSphere [19], using Ubuntu server 22.04-based Virtual Machines (VMs). As can be seen in Figure 1, the deployed Kubernetes cluster has three main components, one master node (8GB RAM, 4vCPUs) and two worker nodes (4GB RAM, 2vCPUs per VM), that were configured and managed using Vagrant [20].



NAME	STATUS	ROLES	AGE	VERSION
controlplane	Ready	control-plane	74m	v1.29.3
node01	Ready	<none>	22m	v1.29.3
node02	Ready	<none>	32s	v1.29.3

Fig. 1. Kubernetes cluster used for testing

1) *Horizontal Pod Autoscaler (HPA)*: Our HPA results, based on the architecture detailed in Figure 2, demonstrate its capability to dynamically scale the number of pods in response to varying network loads. This feature is particularly valuable in managing network traffic spikes, ensuring that the network's performance remains stable despite fluctuating demand. Our study documented several test scenarios where HPA effectively managed the scaling of VNFs, aligning resource allocation with current load conditions without human intervention as can be seen in Figure 3.

2) *Vertical Pod Autoscaler (VPA)*: VPA's role is complementary to that of HPA. While HPA scales the number of instances, VPA adjusts the resources available to each instance. Our test architecture is highlighted in Figure 4 and it shows how VPA could optimize resource usage by allocating more memory and CPU resources to pods under heavy load, thus enhancing the performance of the network functions. The automated adjustment of resources, showcased in Figure 5, helps in maintaining an optimal performance level for the target NF, potentially reducing the need for over-provisioning and consequently, operational costs.

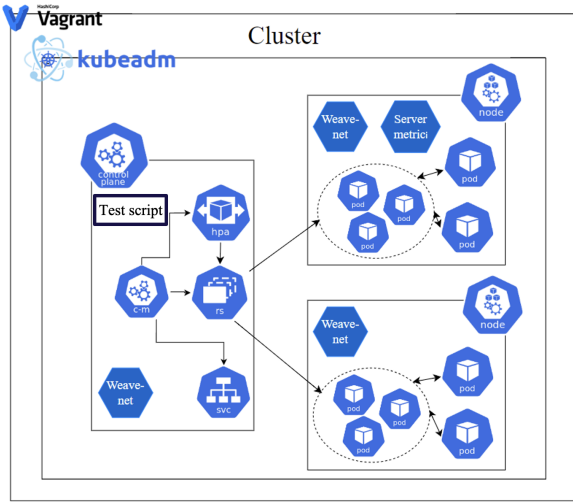


Fig. 2. Kubernetes HPA test architecture

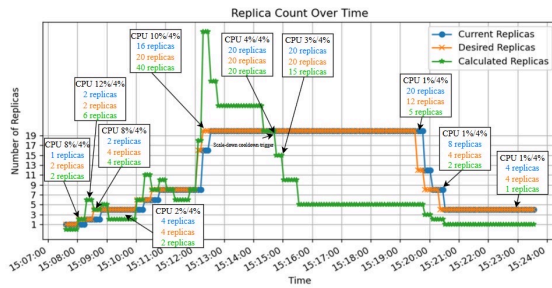


Fig. 3. Kubernetes HPA test results

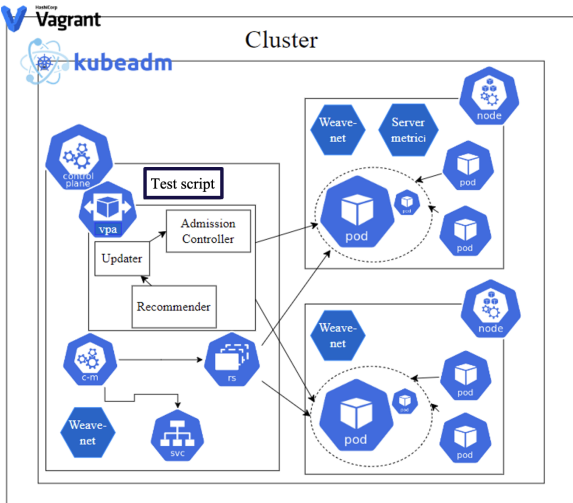


Fig. 4. Kubernetes VPA test architecture

Impact on Network Performance

The integration of HPA and VPA in Kubernetes provides a robust mechanism for handling real-time changes in network demand. The results of our testing indicate that this integration

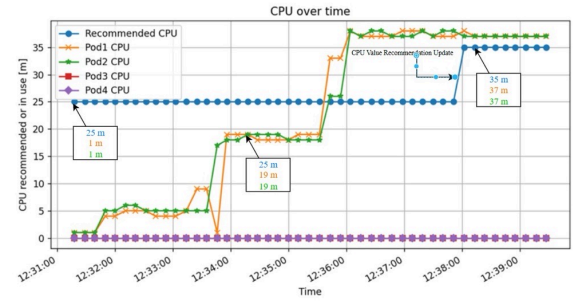


Fig. 5. Kubernetes VPA test results

not only supports scalability but also could improve the reliability and efficiency of network services. The dynamic scaling capabilities of Kubernetes orchestration have shown significant potential in reducing latency and increasing throughput, which are critical metrics for the performance of B5G networks.

Despite the advantages, there are several challenges associated with implementing Kubernetes in a telco environment. The complexity of network configurations, the need for seamless integration with existing network management systems, and the potential for resource contention are notable concerns. Additionally, the latency in scaling operations and the potential for resource wastage during off-peak periods are areas highlighted in our tests and require further refinement.

ANALYSIS OF NWDAF MACHINE LEARNING RESULTS

In this section we explore the detailed results from the implementation of a proprietary Machine Learning-enabled NWDAF based on the open-source implementation provided by OpenAirInterface [21]. The primary focus is on the efficacy of NWDAF in anomaly detection and resource management, which are critical for the proactive maintenance and optimization of network functions in B5G networks.

NWDAF represents a significant advancement in network management technology, being set to integrate Machine Learning capabilities to provide analytics and insights that enhance the operational capabilities of networks. As 5G and future networks move towards more automated and intelligent frameworks, NWDAF plays a pivotal role in enabling these advancements by analyzing network data to predict load levels and identify potential anomalies.

Machine Learning Techniques in NWDAF

According to our findings, NWDAF could utilize various Machine Learning algorithms to analyze network data effectively. These techniques include:

1) *Unsupervised Learning*: NWDAF could apply unsupervised learning algorithms to detect unusual patterns in network traffic without prior labeling of the data. This is particularly useful for identifying new or unexpected types of network anomalies.

2) *Supervised Learning*: Supervised learning algorithms could be employed to predict network loads based on historical data. These models are trained using labeled datasets that represent different traffic conditions, enabling NWDAF to make accurate predictions about future network behaviors.

3) *Semi-supervised Learning*: This approach combines elements of both supervised and unsupervised learning and could be used in situations where only partial data labeling is possible. It is particularly effective in continuously adapting to new data while maintaining the accuracy of the model predictions.

Results from Anomaly Detection and Resource Optimization

Our work illustrates several instances where the proprietary approach to NWDAF was able to successfully detect anomalies and manage resources efficiently. Key findings include:

- The ability of NWDAF to rapidly identify and respond to unexpected spikes in network load on a specific slice, thereby preventing potential service degradation.
- Effective prediction of load levels enabled by ML algorithms, which facilitated proactive resource allocation and optimization, particularly for the UPF CNFs, thus enhancing the overall efficiency and stability of the network.

The summary of our tests, as well as the implemented ML parameters, are highlighted in Table I. Moreover, the results of the three test scenarios are showcased in Figures 6, 7 and 8. As can be observed, the usage of an RMSprop optimizer, together with a larger input dimension, makes the 3rd scenario the best in terms of predicting the slice load level in real-time.

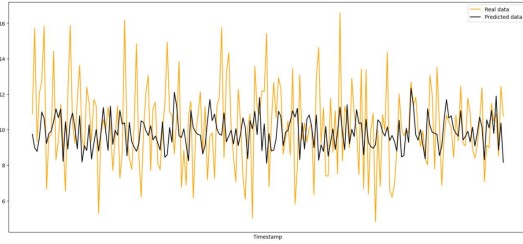


Fig. 6. Slice load anomaly detection - scenario 1

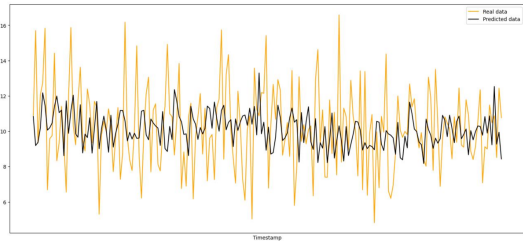


Fig. 7. Slice load anomaly detection - scenario 2

The integration of NWDAF within network management systems could therefore shown a significant positive impact on network performance, as evidenced by the results showcased above. Enhanced anomaly detection and predictive analytics

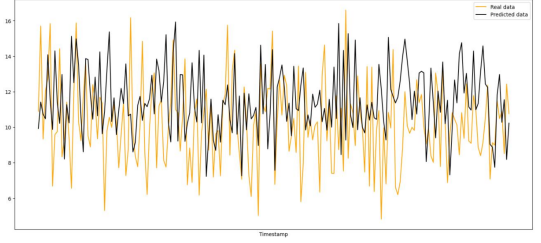


Fig. 8. Slice load anomaly detection - scenario 3

contribute to more reliable and consistent network performance, with reduced downtime and improved user satisfaction if combined with advanced resource scaling capabilities.

Despite the promising results, several challenges were noted in the deployment and operation of the emulated NWDAF. These include the complexity of training ML models on large-scale network data, the need for continuous model updates to adapt to changing network conditions, and ensuring data privacy and security during the analytics process.

Thus, the analysis of NWDAF's Machine Learning results underscores its potential to transform network management through enhanced intelligence and automation. Future developments should focus on refining ML models, improving the scalability of NWDAF implementations, and integrating more advanced AI techniques to further enhance predictive accuracy and operational efficiency.

EXPERIMENTATION AND RESULTS IN AN END-TO-END SCENARIO

In this section we present the experimental setup and results from deploying Kubernetes orchestrated 5GC CNFs and the NWDAF within our pre-commercial testbed environment. The main focus is on assessing the load management capabilities of this End-to-End (E2E) system under simulated network conditions.

Experimental Setup

The E2E experimental 5G Standalone (SA) testbed was configured using open-source software components: OpenAir-Interface (OAI) as the 5G Core Network (5GC), including the NWDAF features [21], and gnb-sim [22] as the 5G Radio Access Network (RAN) and User Equipment (UE) emulator. We've chosen this setup to simulate a realistic 5G network environment where both data traffic and network functions could be controlled and monitored.

We've leveraged on the previously mentioned Kubernetes cluster to manage the OAI 5GC and NWDAF CNFs, including the capability to choose between the HPA and VPA options to dynamically adjust compute resources, specifically for the UPF CNFs, based on slice-level network demand. Additionally, NWDAF was integrated to utilize the proprietary developed ML capabilities for anomaly detection and resource prediction at the network slice level.

The high-level 5G SA network architecture is showcased in Figure 9. The purpose of the E2E system is highlighted in

TABLE I

Testing nr.	Input dim	Hidden dim	Latent dim	Loss function	Learning rate	Epochs	Optimizer
1	100	30	5	MSELoss	1e-2	1	Adam
2	100	50	10	MSELoss	1e-2	2	Adam
3	400	25	5	MSELoss	1e-2	2	RMSprop

Figure 10 highlighting the usage of the NWDAF and a proprietary Decision Making Engine block to scale the number of UPF pods depending on the network-slice level load predicted by the pre-trained ML model.

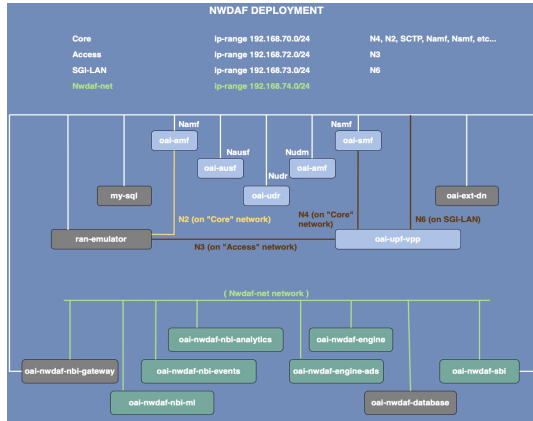


Fig. 9. Simulated 5G SA network architecture [21]

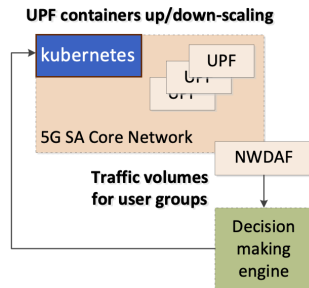


Fig. 10. E2E system architecture

Furthermore, the logic embedded into the Decision Making Engine follows the principles described in Algorithm 1.

Performance Benchmarks, Results and Future Directions

The network's performance was evaluated under various emulated slice-level load scenarios to assess the impact of dynamic scaling and ML-based predictions on network behavior:

- **Throughput and Latency:** Measurements were taken to determine the network throughput and latency during normal operation and under stress tests that simulated high traffic conditions.
- **Resource Utilization:** CPU and memory usage were monitored to evaluate the efficiency of resource usage by Kubernetes and the effectiveness of the two existing

Algorithm 1 UPF pods scaling based on ML predictions

```

function FETCH_DATA
    result  $\leftarrow$  slice-level load data from NWDAF
    if result.returncode  $\neq$  0 then
        SYS.EXIT(1)
    end if
    return JSON.LOADS(result.stdout)
end function

function SCALE_PODS(scale_type)
    command  $\leftarrow$  Kubernetes API for HPA/VPA depending on
scale_type
end function

procedure MAIN
    TIMER(1.0, main).start()
    data  $\leftarrow$  FETCH_DATA
    prediction  $\leftarrow$  ML_MODEL(data)
    if prediction = high_traffic then SCALE_PODS(up)
    else if prediction = low_traffic then SCALE_PODS(down)
    end if
end procedure

```

autoscaling options (HPA/VPA) in optimizing resource allocations.

In terms of the actual benefits of our solution, below we have a high-level summary of what could be achieved in a real-world environment, but still needs to be validated in our future work:

- **Enhanced Network Performance:** The introduction of NWDAF-based ML optimizations should led to significant improvements in network efficiency, with reduced latency and higher throughput observed during peak load times.
- **Energy Efficiency:** AI-driven resource management should help reduce energy consumption by minimizing the resources during low traffic while adequately scaling up during high demand.
- **Effective Load Management:** The dynamic scaling provided by Kubernetes and the predictive analytics from NWDAF should enable the network to handle sudden traffic spikes more efficiently than the static setup.

These testing scenarios and objectives should demonstrate the potential of integrating Kubernetes and NWDAF in managing and optimizing novel 6G network operations. AI-driven optimizations not only could enhance network performance but also should contribute to significant energy savings and improved load management.

Looking forward, the research points towards several future

directions for demonstrating the efficacy and applicability of AI-driven network management in real-world scenarios:

- **Validating the E2E Setup in a Real-World Environment:** Implementing the developed components into our Future Networks lab industrial-grade 5G network setup to assess its performance characteristics.
- **Integration with Emerging Technologies:** Further research could explore the integration of NWDAF and Kubernetes with emerging technologies like edge computing and to optimize the placement of network applications in different computing domains depending on the load location.
- **Advanced Predictive Analytics:** Implementing more sophisticated ML and potentially Deep Learning algorithms could provide even more accurate predictions and automation within network operations.

CONCLUSION

Our research explored the integration of AI-driven Zero-Touch technologies, particularly ML-based NWDAF, in CNFs scaling automation for novel 6G networks. The findings indicate that Kubernetes can effectively orchestrate CNFs, enhancing scalability and resource management through dynamic resource allocation. NWDAF-enabled Machine Learning algorithms can facilitate proactive anomaly detection and predictive resource management, optimizing network performance and reducing potential disruptions based on the proprietary developed Decision Making Engine component.

The adoption of these technologies in 6G network management could significantly improve how network services are delivered and maintained. Automating complex processes and predicting network requirements in real-time will support the advanced applications envisioned for 6G, such as integrated sensing and communications, enhanced ultra-reliable low-latency communications, and massive IoT communications for future Digital Twins. However, the complexity of implementation, compatibility issues, and scalability concerns pose challenges that must be addressed to fully realize these benefits.

Our future research will focus on validating the performance of our E2E setup, integrating more advanced AI techniques to enhance the predictive accuracy and decision-making capabilities within network management systems. Investigating cross-domain optimizations and developing robust security mechanisms to protect network data are also crucial. The potential of AI-driven technologies to revolutionize network management promises not only enhanced operational efficiency but also demands for innovation in service offerings, propelling us towards a more connected and efficient digital society.

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